

AN IMPROVED UPPER BOUND FOR LITTLEWOOD'S PROBLEM ON ZEROS OF COSINE POLYNOMIALS

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ABSTRACT. For a finite set $A \subset \mathbb{Z}_{\geq 0}$ of size N , let $F_A(x) = \sum_{a \in A} \cos(ax)$ and let $Z(N)$ denote the minimum, over all such A , of the number of distinct zeros of F_A in $[0, 2\pi)$. Determining the order of $Z(N)$ is Littlewood's problem on the least number of zeros of a cosine polynomial. The best known upper bound, due to Konyagin and to Juškevičius and Sahasrabudhe, is $Z(N) \ll N^{2/3}(\log N)^{2/3}$. We prove that for all sufficiently large N ,

$$Z(N) \ll N^{2/3}(\log N)^{1/3},$$

improving the logarithmic factor by $(\log N)^{1/3}$. The construction realises F_A as $D_n - g$ for a random 0/1 cosine polynomial g built from independently placed contiguous frequency blocks; non-resonant phase averaging and a weighted Littlewood–Offord inequality force the small-band set where $|D_n - g|$ can vanish to have expected measure $\ll \sqrt{(\log m)/m}$, while a deterministic argument bounds the zeros of $D_n - g$ in terms of that measure.

1. INTRODUCTION

The problem. For a finite set $A \subset \mathbb{Z}_{\geq 0}$ write

$$F_A(x) = \sum_{a \in A} \cos(ax), \quad x \in [0, 2\pi),$$

a real cosine polynomial with 0/1 coefficients. For a cosine polynomial f let $Z(f)$ denote the number of distinct zeros of f in $[0, 2\pi)$, each zero counted once as a point of the circle irrespective of its multiplicity, and set

$$Z(N) = \min_{|A|=N} Z(F_A).$$

The question of how small $Z(N)$ can be is Littlewood's problem on the least number of real zeros of a cosine polynomial with a prescribed number of terms (Problem 22 in his list). The boundary cases are $Z(1) = 0$ and $Z(2) = 1$, while any nonzero cosine polynomial of degree d has at most $2d$ zeros per period; the difficulty lies entirely in the lower range, where one seeks sets A whose cosine polynomial oscillates as little as possible while carrying N frequencies. The problem sits at the interface of harmonic analysis and combinatorics: the frequencies of A are unrestricted in size, so extremal configurations must exploit arithmetic structure rather than degree.

Previous work. In 2008, Borwein, Erdélyi, Ferguson and Lockhart disproved Littlewood's near-linear suggestion by constructing examples with $O(N^{5/6} \log N)$ zeros [1]. The qualitative assertion $Z(N) \rightarrow \infty$ was later proved independently by Erdélyi [2] and Sahasrabudhe [3]. The latter obtained the quantitative lower bound $(\log \log \log N)^{1/2-o(1)}$, which Erdélyi subsequently improved to $(\log \log \log N)^{1-o(1)}$ [4]. On the upper side, the sharpest bound prior to the present work is

$$Z(N) \ll N^{2/3}(\log N)^{2/3},$$

proved by Konyagin [5] and, independently and by a different route, by Juškevičius and Sa-hasrabudhe [6]. In the opposite direction, Bedert's current lower bound is

$$Z(N) \gg \frac{\log \log N}{\log \log \log N} = (\log \log N)^{1-o(1)}$$

for sufficiently large N [7], so the true order of $Z(N)$ is not yet known even up to a power of N . Our contribution lies on the upper side.

Main result.

Theorem 1. *For all sufficiently large N ,*

$$Z(N) \ll N^{2/3}(\log N)^{1/3}.$$

This improves the logarithmic factor in the previously known upper bound by $(\log N)^{1/3}$; the power $N^{2/3}$ is unchanged. The set A realising the bound is obtained probabilistically: an averaging argument yields a suitable 0/1 cosine polynomial g , and A is then the frequency support of $D_n - g$, where $D_n(x) = \sum_{k=0}^n \cos(kx)$ is the Dirichlet kernel.

Strategy of the proof. The proof rests on two independent components, one probabilistic and one deterministic, joined through the small-band set. Write $D_n(x) = \frac{1}{2} + \sin(Mx)/(2 \sin(x/2))$ with $M = n + \frac{1}{2}$, and for a 0/1 cosine polynomial g of degree at most $m < n$ put $h(x) = 2 \sin(x/2)(g(x) - \frac{1}{2})$. Then $D_n(x) - g(x) = 0$ is equivalent to $\sin(Mx) = h(x)$, and the zeros of $D_n - g$ can occur only on the *small-band set*

$$E(g) = \{x \in (0, \pi) : |h(x)| \leq 1\},$$

where the graph of h meets the strip in which $\sin(Mx)$ takes its values.

The first component (Innovation A) constructs a random g for which $E(g)$ is typically very thin. We partition the frequency axis into q boxes of length $4L$ and, in each box independently, select a contiguous block I_j of a random length $R_j \in \{L, \dots, 2L - 1\}$ at a random offset $U_j \in \{0, \dots, L - 1\}$, retaining the block with an independent fair coin η_j . The exact evaluation of a block sum shows that its size is governed by the scale $A_L(x) = \min\{L, 1/\sin(x/2)\}$. Away from a thin resonance layer $R = \{x : |\sin(4Lx)| \leq D/q\}$, a non-resonant phase average forces a positive proportion of the box phases to be well separated from the axis; combined with a length-energy estimate for the block lengths, this yields a fixed positive proportion of triples (j, u, r) whose blocks are of order $A_L(x)$, and hence, with probability $1 - e^{-cq}$, a linear number $\gg q$ of retained blocks with coefficient of size $\gg A_L(x)$. For such an x , the event $x \in E(g)$ reads $|\sum_j \eta_j b_j(x) - \frac{1}{2}| \leq 1/(2 \sin(x/2))$, an anti-concentration event for a signed sum with $\gg q$ weights of size $\gg A_L(x)$. A weighted Littlewood-Offord inequality with an arbitrary centre, applied after conditioning on the box geometry, bounds its probability by $\ll (1 + w/a)/\sqrt{q}$. Integrating the resulting pointwise bounds in x — the resonance layer contributing $O(1/q)$, and the range $0 < x \leq 1/L$ contributing $\int_{(L\sqrt{q})^{-1}}^{1/L} dx/(xL\sqrt{q}) \ll (\log \sqrt{q})/(L\sqrt{q})$ through the small- x ratio $w/a \asymp 1/(Lx)$ — gives $\mathbb{E}|E(g)| \ll 1/q + (\log q)/(L\sqrt{q}) + 1/\sqrt{q}$. With $q \asymp m/L$ and $L \asymp \log m$ this is $\ll \sqrt{(\log m)/m}$, and the averaging principle produces a single deterministic g attaining the bound.

The second component (Innovation B) is a deterministic estimate for the number of zeros of $D_n - g$ in terms of $|E(g)|$ alone:

$$Z(D_n - g) \ll n|E(g)| + m.$$

On $E(g)$ one sets $\alpha(x) = \arcsin h(x)$, so that $\sin(Mx) = h(x)$ is equivalent to $Mx - \alpha(x) \in 2\pi\mathbb{Z}$ or $Mx + \alpha(x) \in (2\mathbb{Z} + 1)\pi$. Each phase $\psi_{\pm} = Mx \pm \alpha$ is treated on the intervals of monotonicity

determined by the zeros of the auxiliary polynomial $P = M^2(1 - h^2) - (h')^2$, which controls the interior critical points of both phases and which, by $|h'(0)| \leq m + \frac{1}{2} < M$ (here the hypothesis $n > m$ is used), is not identically zero and has $O(m)$ zeros. A strictly monotone phase meets a lattice of spacing 2π at most $1 + \text{Var}/(2\pi)$ times, and the total variation of $\psi_\pm$ over $E(g)$ is $\leq M|E(g)| + \text{Var}(\alpha)$ with $\text{Var}(\alpha) = O(m)$; summing over the $O(m)$ intervals gives the stated count.

Combining the two components, one fixes g from the probabilistic construction, sets $n = N + K - 1$ where K is the number of nonzero coefficients of g , so that $D_n - g$ has exactly N nonzero $0/1$ coefficients, and chooses $m = \lceil N^{2/3}(\log N)^{1/3} \rceil$, $L = \lfloor \log m \rfloor$. Then $n \asymp N$, $|E(g)| \ll \sqrt{(\log m)/m}$, and the deterministic count yields $Z(N) \ll n\sqrt{(\log m)/m} + m \ll N^{2/3}(\log N)^{1/3}$.

Organisation. Section 2 records the Dirichlet reduction and introduces the small-band set. Section 3 describes the random block construction and establishes the exact block-sum identity and the length–energy estimate. Section 4 treats the resonance layer and the non-resonant phase average, from which the proportion of good triples and the abundance of large coefficients follow. Section 5 proves the weighted Littlewood–Offord inequality and deduces the pointwise small-ball bound, and Section 6 integrates it to bound $\mathbb{E}|E(g)|$ and to extract a deterministic g . Section 7 gives the deterministic zero count, and Section 8 balances the parameters, fixes the exact term count, and completes the proof of Theorem 1.

2. PRELIMINARIES AND THE DIRICHLET-KERNEL REDUCTION

Notation and conventions. Throughout, x denotes a real variable, and we write $s = s(x) = \sin(x/2)$. For real-valued quantities f and $g \geq 0$ we write $f \ll g$ (Vinogradov) if $|f| \leq Cg$ for an absolute constant C , and $f \asymp g$ if $f \ll g \ll f$; all implied constants are absolute unless a dependence is indicated. The notation $O(g)$ carries the same meaning as $\ll g$. Logarithms are natural. For a set S we write $|S|$ for its Lebesgue measure when $S \subseteq \mathbb{R}$ and for its cardinality when S is finite; the two usages are always distinguished by context.

For $L \geq 1$ we set

$$A_L(x) = \min\{L, 1/s\}, \quad \Delta(x) = \min\{Ls, 1\},$$

the natural block scale and its normalized companion; note $\Delta = Ls A_L$ when $s > 0$, and A_L, Δ are used only for $0 < x < 2\pi$, where $s > 0$.

Given a real trigonometric polynomial f , its zeros are counted as *distinct points*: $Z(f)$ denotes the number of distinct $x \in [0, 2\pi)$ with $f(x) = 0$. For $A \subseteq \mathbb{Z}_{\geq 0}$ with $|A| = N$ we write $F_A(x) = \sum_{a \in A} \cos(ax)$, and

$$Z(N) = \min_{|A|=N} Z(F_A),$$

the minimal number of distinct zeros in $[0, 2\pi)$ over all frequency sets of cardinality N .

The Dirichlet kernel and the reduction. Write

$$D_n(x) = \sum_{k=0}^n \cos(kx)$$

for the Dirichlet cosine kernel of order n .

Lemma 1 (Dirichlet identity). *For every $x \notin 2\pi\mathbb{Z}$,*

$$D_n(x) = \frac{1}{2} + \frac{\sin((n + \frac{1}{2})x)}{2\sin(x/2)}.$$

Proof. For each integer $k \geq 0$ the product-to-sum identity gives

$$2 \sin(x/2) \cos(kx) = \sin((k + \frac{1}{2})x) - \sin((k - \frac{1}{2})x).$$

Summing over $k = 0, 1, \dots, n$, the right-hand side telescopes to

$$2 \sin(x/2) D_n(x) = \sin((n + \frac{1}{2})x) + \sin(x/2).$$

For $x \notin 2\pi\mathbb{Z}$ we have $\sin(x/2) \neq 0$, and dividing by $2 \sin(x/2)$ yields the stated identity. $\square$

We now fix the objects that carry the reduction. Let g be a cosine polynomial with coefficients in $\{0, 1\}$ and degree at most m , and let $n > m$. Set

$$M = n + \frac{1}{2}, \quad h(x) = 2 \sin(x/2) (g(x) - \frac{1}{2}),$$

so that h is a real trigonometric polynomial with frequencies in $\frac{1}{2} + \mathbb{Z}$; its top frequency does not exceed $m + \frac{1}{2}$. Since $g(0) = |\{k : g_k = 1\}|$ is finite and $\sin(0) = 0$, we have $h(0) = 0$.

Lemma 2 (Dirichlet-kernel reduction). *Let g be a 0/1 cosine polynomial of degree at most m and let $n > m$, with M and h as above. Then for every $x \in (0, 2\pi)$,*

$$D_n(x) - g(x) = 0 \iff \sin(Mx) = h(x).$$

In particular, every zero of $D_n - g$ in $(0, 2\pi)$ satisfies $|h(x)| \leq 1$.

Proof. Fix $x \in (0, 2\pi)$, so that $x \notin 2\pi\mathbb{Z}$ and $\sin(x/2) \neq 0$. By Lemma 1,

$$D_n(x) - g(x) = \frac{\sin(Mx)}{2 \sin(x/2)} + \frac{1}{2} - g(x) = \frac{1}{2 \sin(x/2)} \left(\sin(Mx) - 2 \sin(x/2) (g(x) - \frac{1}{2}) \right) = \frac{\sin(Mx) - h(x)}{2 \sin(x/2)}.$$

As $2 \sin(x/2) \neq 0$, the left-hand side vanishes precisely when $\sin(Mx) = h(x)$. The final assertion follows since $|\sin(Mx)| \leq 1$. $\square$

The equivalence localizes the analysis to the region where $\sin(Mx)$ can meet h . Accordingly we introduce the *small-band set*

$$E(g) = \{x \in (0, \pi) : |h(x)| \leq 1\}.$$

By Lemma 2, membership in $E(g)$ is a *necessary* condition for a point of $(0, \pi)$ to be a zero of $D_n - g$: no zero of $D_n - g$ occurs at a point of $(0, \pi)$ where $|h| > 1$. We stress that this is a one-sided characterization; points of $E(g)$ need not be zeros, and the equivalence of Lemma 2 is the tool by which zeros will later be counted.

We confine $E(g)$ to the half-open half-period $(0, \pi)$. The kernel D_n and any 0/1 cosine polynomial g are even and 2π -periodic, so $D_n - g$ takes the same values at x and at $2\pi - x$; this symmetry lets one pass from a count on $(0, \pi)$ to a count on the full period $[0, 2\pi)$ at the cost of endpoint contributions only. We defer this full-period/half-period passage, together with the treatment of the endpoints $x \in \{0, \pi\}$, to the deterministic zero-counting section, where the number of zeros of $D_n - g$ is estimated from $|E(g)|$ and the analytic structure of h .

3. THE RANDOM BLOCK CONSTRUCTION

Fix a large integer m , and an integer L with

$$8 \leq L \leq m^{1/3}.$$

Set

$$q = \left\lfloor \frac{m+1}{4L} \right\rfloor,$$

and partition the frequency axis into q consecutive boxes, the j -th being the set of integers in the interval of length $4L$ beginning at $4Lj$, for $j \in \{0, \dots, q-1\}$. Within each box we select a random contiguous block of frequencies, as follows.

For each $j \in \{0, \dots, q-1\}$, independently, let

$$U_j \sim \text{Unif}\{0, \dots, L-1\}, \quad R_j \sim \text{Unif}\{L, \dots, 2L-1\},$$

and put

$$I_j = \{4Lj + U_j, 4Lj + U_j + 1, \dots, 4Lj + U_j + R_j - 1\}.$$

Thus I_j is a block of R_j consecutive integers starting at $4Lj + U_j$. Since

$$U_j + R_j - 1 \leq (L-1) + (2L-1) - 1 = 3L-2 < 4L,$$

the block I_j is contained in the j -th box; in particular the blocks $I_0, \dots, I_{q-1}$ are pairwise disjoint. Finally let $\eta_0, \dots, \eta_{q-1}$ be independent Bernoulli(1/2) random variables, independent of the U_j and R_j , and set

$$g(x) = \sum_{j=0}^{q-1} \eta_j \sum_{k \in I_j} \cos(kx).$$

Lemma 3. *The trigonometric polynomial g has all coefficients in $\{0, 1\}$, and $\deg g \leq m$.*

Proof. The blocks I_j are pairwise disjoint, so each frequency k occurs in at most one inner sum; its coefficient is $\eta_j \in \{0, 1\}$ when $k \in I_j$ and 0 otherwise. For the degree, every frequency appearing in g lies in some I_j , and the largest such frequency is at most the right endpoint of the last block I_{q-1} . That endpoint is bounded by

$$4L(q-1) + U_{q-1} + R_{q-1} - 1 \leq 4L(q-1) + (L-1) + (2L-1) - 1 = 4Lq - L - 3.$$

By the definition of q we have $4Lq \leq m+1$, whence

$$4Lq - L - 3 \leq (m+1) - L - 3 = m - (L+2) \leq m,$$

using $L \geq 8$. Therefore $\deg g \leq m$. $\square$

It is convenient to record the contribution of a single contiguous block in closed form. For integers $j \geq 0$, $u \geq 0$ and $r \geq 1$ write

$$b_{j,u,r}(x) = \sum_{k=4Lj+u}^{4Lj+u+r-1} \cos(kx),$$

so that $g(x) = \sum_j \eta_j b_{j,U_j,R_j}(x)$. Throughout we abbreviate $s = \sin(x/2)$.

Lemma 4 (Exact block sum). *For $0 < x < 2\pi$ and all integers $j \geq 0$, $u \geq 0$, $r \geq 1$,*

$$2s b_{j,u,r}(x) = 2 \sin\left(\frac{rx}{2}\right) \cos\left(4Ljx + \frac{(2u+r-1)x}{2}\right).$$

Proof. Write $a = 4Lj + u$ for the initial frequency of the block. Summing the finite geometric series with ratio $e^{ix} \neq 1$,

$$\sum_{k=a}^{a+r-1} e^{ikx} = e^{iax} \frac{e^{irx} - 1}{e^{ix} - 1} = e^{i(a+\frac{r-1}{2})x} \frac{\sin(rx/2)}{\sin(x/2)},$$

where in the last step we have factored $e^{irx/2}$ from the numerator and $e^{ix/2}$ from the denominator. Taking real parts and using $\sin(x/2) = s$,

$$b_{j,u,r}(x) = \frac{\sin(rx/2)}{s} \cos\left(\left(a + \frac{r-1}{2}\right)x\right).$$

Since $a + \frac{r-1}{2} = 4Lj + u + \frac{r-1}{2} = 4Lj + \frac{2u+r-1}{2}$, multiplying by $2s$ gives the stated identity. $\square$

The factor $\sin(rx/2)$ in Lemma 4 varies over $r \in \{L, \dots, 2L-1\}$, and its typical size fixes the natural scale of a block. Accordingly we introduce the *block scale*

$$A_L(x) = \min \left\{ L, \frac{1}{s} \right\},$$

which will govern the size of $|b_{j,u,r}(x)|$ for a positive proportion of blocks in the analysis that follows.

4. BLOCK ENERGY AND NON-RESONANCE

Throughout this section we work on the half-period $(0, \pi)$ and write $s = \sin(x/2)$. Recall the natural block scale $A_L(x) = \min\{L, 1/s\}$ and $\Delta(x) = \min\{Ls, 1\}$. The two mechanisms developed here are complementary: a length–energy estimate showing that, on average over the admissible block lengths, a block carries energy comparable to Δ^2 , and a phase analysis showing that away from a small resonance layer the box centres are equidistributed enough to convert this energy into a genuine per-box contribution.

4.1. A length–energy estimate.

Lemma 5 (Contiguous block-length square-mean). *There is an absolute constant $c_A > 0$ such that for all integers $L \geq 8$ and all t with $0 < t \leq \pi/2$,*

$$\frac{1}{L} \sum_{r=L}^{2L-1} \sin^2(rt) \geq c_A \min\{(L \sin t)^2, 1\}.$$

One may take $c_A = 10^{-3}$.

Proof. Writing $\sin^2 y = \frac{1}{2}(1 - \cos 2y)$ and summing the resulting geometric progression of cosines over $r = L, \dots, 2L-1$ gives the exact identity

$$\frac{1}{L} \sum_{r=L}^{2L-1} \sin^2(rt) = \frac{1}{2} - \frac{\sin(Lt)}{2L \sin t} \cos((3L-1)t). \quad (1)$$

Indeed, $\sum_{r=L}^{2L-1} \cos(2rt) = \operatorname{Re} \sum_{r=L}^{2L-1} e^{2irt}$, and evaluating the finite geometric sum yields $\sum_{r=L}^{2L-1} \cos(2rt) = \frac{\sin(Lt)}{\sin t} \cos((3L-1)t)$, whence (1).

We distinguish three cases.

Case 1: $L \sin t \geq 2$. Here $\min\{(L \sin t)^2, 1\} = 1$, and the second term in (1) is bounded in absolute value by $\frac{1}{2L \sin t} \leq \frac{1}{4}$. Hence the left-hand side of (1) is at least $\frac{1}{2} - \frac{1}{4} = \frac{1}{4} \geq c_A$.

In the remaining two cases $L \sin t < 2$, so $\min\{(L \sin t)^2, 1\} = (L \sin t)^2$. Since $\sin t \geq 2t/\pi$ on $(0, \pi/2]$, we have $Lt \leq \frac{\pi}{2} L \sin t < \pi$.

Case 2: $Lt \leq \pi/4$. For each r with $L \leq r \leq 2L-1$ we have $0 < rt \leq (2L-1)t < 2Lt \leq \pi/2$, so $\sin(rt) \geq 2rt/\pi \geq 2Lt/\pi$ by concavity of $\sin$ on $[0, \pi/2]$. Therefore

$$\frac{1}{L} \sum_{r=L}^{2L-1} \sin^2(rt) \geq \frac{4}{\pi^2} L^2 t^2 \geq \frac{4}{\pi^2} (L \sin t)^2,$$

using $\sin t \leq t$.

Case 3: $\pi/4 < Lt < \pi$. We bound the second term of (1) through the quotient $\phi(t) = \sin(Lt)/(L \sin t)$, considered on $(0, \pi/L)$, where $\sin(Lt) > 0$.

We first show that ϕ is nonincreasing on $(0, \pi/L)$. Write $\phi(t) = \psi(Lt)/\psi(t)$, where $\psi(u) = \sin u/u$ for $u \neq 0$ and $\psi(0) = 1$; indeed $\phi(t) = \frac{\sin(Lt)}{L \sin t} = \frac{\sin(Lt)/(Lt)}{\sin t/t} = \frac{\psi(Lt)}{\psi(t)}$. Taking logarithmic derivatives,

$$\frac{\phi'(t)}{\phi(t)} = L \frac{\psi'(Lt)}{\psi(Lt)} - \frac{\psi'(t)}{\psi(t)} = \frac{1}{t} \left(F(Lt) - F(t) \right), \quad F(u) := u \frac{\psi'(u)}{\psi(u)} = u \cot u - 1.$$

The map $u \mapsto u \cot u$ is strictly decreasing on $(0, \pi)$, since its derivative is $\cot u - u \csc^2 u = (\sin u \cos u - u)/\sin^2 u < 0$ there (as $\sin u \cos u = \frac{1}{2} \sin 2u < u$ for $0 < u < \pi$). Hence F is strictly decreasing on $(0, \pi)$, and for $0 < t < \pi/L$ we have $t < Lt < \pi$, so $F(Lt) < F(t)$ and $\phi'(t) \leq 0$. Thus ϕ is nonincreasing on $(0, \pi/L)$, and for $t > \pi/(4L)$ (equivalently $Lt > \pi/4$),

$$0 \leq \frac{\sin(Lt)}{L \sin t} = \phi(t) \leq \phi\left(\frac{\pi}{4L}\right) = \frac{\sin(\pi/4)}{L \sin(\pi/(4L))}.$$

It remains to bound the right-hand side uniformly for $L \geq 8$. Put $u = \pi/(4L) \in (0, \pi/32]$ and $G(L) := L \sin(\pi/(4L)) = \frac{\pi}{4} \psi(u)$. Since $\psi(u) = \sin u/u$ is decreasing on $(0, \pi)$ and $u = \pi/(4L)$ decreases as L increases, $\psi(u)$ increases with L ; hence G is increasing in L , and

$$\frac{\sin(\pi/4)}{L \sin(\pi/(4L))} = \frac{\sin(\pi/4)}{G(L)} \text{ is decreasing in } L.$$

Its supremum over $L \geq 8$ is therefore attained at $L = 8$, giving

$$\frac{\sin(\pi/4)}{L \sin(\pi/(4L))} \leq \frac{\sin(\pi/4)}{8 \sin(\pi/32)} = 0.90177 \dots < \cos(\pi/8) = 0.92388 \dots \quad (L \geq 8).$$

(As $L \rightarrow \infty$ the quotient decreases to $\sin(\pi/4)/(\pi/4) = 2\sqrt{2}/\pi = 0.90031 \dots$, so the displayed bound is in fact uniform.) Consequently, for $\pi/4 < Lt < \pi$ and $L \geq 8$,

$$\frac{1}{L} \sum_{r=L}^{2L-1} \sin^2(rt) \geq \frac{1}{2} - \frac{1}{2} \cdot \frac{\sin(\pi/4)}{8 \sin(\pi/32)} > \frac{1 - \cos(\pi/8)}{2} > 0.038.$$

Since $L \sin t < 2$ forces $(L \sin t)^2 < 4$, the bound $0.038 > c_A \cdot 4$ holds for the value of c_A below.

Combining the three cases, the left-hand side is at least $\min\{\frac{1}{4}, \frac{4}{\pi^2}(L \sin t)^2, 0.038\}$ in the respective regimes, and in each regime this exceeds $c_A \min\{(L \sin t)^2, 1\}$ with $c_A = 10^{-3}$. $\square$

4.2. The resonance layer. Fix $D = 4$ and assume $q \geq 8$. The box centres are governed by the frequency $4L$, and we isolate the frequencies at which $\sin(4Lx)$ is too small to guarantee equidistribution of the box phases.

Definition 1. The *resonance layer* is

$$R = \{x \in [0, \pi] : |\sin(4Lx)| \leq D/q\}.$$

Remark 1. The resonance layer must be discarded rather than controlled by a uniform per-block bound. A per-block lower bound carrying an arbitrary extra phase θ fails: at $\theta = \pi/2$ and $x \rightarrow 0$ with $Lx \rightarrow 0$ the block contribution is of order $(Lx)^4$, whereas the energy argument only guarantees an amount of order $(Lx)^2$. It is precisely this loss on R that the phase average below avoids by excluding R .

Lemma 6 (Resonance measure). *For all $q \geq 1$,*

$$|R| = 2 \arcsin(\min\{1, D/q\}) \leq \frac{\pi D}{q},$$

and if $q \geq 2D$ then $|R| = 2 \arcsin(D/q) \leq 4D/q$.

Proof. As x runs over $[0, \pi]$, the argument $y = 4Lx$ runs over $[0, 4L\pi]$, that is, over $4L$ full periods of $|\sin|$. Within a single period the set $\{|\sin y| \leq \delta\}$ has measure $2 \arcsin(\min\{1, \delta\})$; summing over the $4L$ periods and dividing by the Jacobian $4L$ shows that $\{x \in [0, \pi] : |\sin(4Lx)| \leq \delta\}$ has measure $2 \arcsin(\min\{1, \delta\})$, with $\delta = D/q$. The truncation $\min\{1, D/q\}$ is required because $2 \arcsin(D/q)$ is only defined when $D/q \leq 1$. Finally $\arcsin z \leq \frac{\pi}{2}z$ for $0 \leq z \leq 1$ gives the first bound, and $\arcsin z \leq 2z$ for $0 \leq z \leq \frac{1}{2}$ gives the second when $q \geq 2D$. $\square$

Lemma 7 (Non-resonant phase average). *If $x \in [0, \pi] \setminus R$, then for every real α ,*

$$\frac{1}{q} \sum_{j=0}^{q-1} \cos^2(4Ljx + \alpha) \geq \frac{3}{8}.$$

Proof. Using $\cos^2 y = \frac{1}{2}(1 + \cos 2y)$,

$$\frac{1}{q} \sum_{j=0}^{q-1} \cos^2(4Ljx + \alpha) = \frac{1}{2} + \frac{1}{2q} \operatorname{Re} \left(e^{2i\alpha} \sum_{j=0}^{q-1} e^{8iLjx} \right) = \frac{1}{2} + \frac{1}{2q} \operatorname{Re} \left(e^{2i\alpha} \frac{1 - e^{8iLqx}}{1 - e^{8iLx}} \right).$$

The geometric sum is bounded by $|1 - e^{8iLqx}|/|1 - e^{8iLx}| \leq 1/|\sin(4Lx)|$. Since $x \notin R$ we have $|\sin(4Lx)| > D/q$, so the error term is at most

$$\frac{1}{2q|\sin(4Lx)|} < \frac{1}{2D} = \frac{1}{8}.$$

Hence the average is at least $\frac{1}{2} - \frac{1}{8} = \frac{3}{8}$. $\square$

5. ANTI-CONCENTRATION

Throughout this section we retain the notation of the preceding sections. In particular $s = \sin(x/2)$, $\Delta(x) = \min\{Ls, 1\}$, $A_L(x) = \min\{L, 1/s\}$, and $R = \{x \in [0, \pi] : |\sin(4Lx)| \leq D/q\}$ with $D = 4$ and $q \geq 8$. For a triple (j, u, r) with $j \in \{0, \dots, q-1\}$, $u \in \{0, \dots, L-1\}$, $r \in \{L, \dots, 2L-1\}$ we write

$$Y_{j,u,r}(x) = 2s b_{j,u,r}(x) = 2 \sin\left(\frac{rx}{2}\right) \cos\left(4Ljx + \frac{(2u+r-1)x}{2}\right),$$

the last equality being the exact block-sum identity. The aim of the section is to show that, off the resonance layer R , the random polynomial g concentrates in the small-band set $E(g)$ only on a set of small measure. This rests on four ingredients: a fixed proportion of triples on which $|b_{j,u,r}|$ is comparable to $A_L(x)$; an abundance of boxes carrying such a block after the random choice of geometry; a weighted Littlewood–Offord inequality with an arbitrary centre; and the resulting pointwise small-ball bound.

5.1. A fixed proportion of good triples.

Lemma 8 (Fixed proportion of good triples). *There exist absolute constants $c_1, \gamma > 0$ such that for every $x \in (0, \pi) \setminus R$ at least a proportion γ of the qL^2 triples (j, u, r) satisfy*

$$|b_{j,u,r}(x)| \geq c_1 A_L(x).$$

Proof. Fix $x \in (0, \pi) \setminus R$ and write $t = x/2 \in (0, \pi/2)$, so that $s = \sin t$ and $\Delta = \min\{Ls, 1\}$.

Second-moment lower bound. By the block-sum identity,

$$Y_{j,u,r}(x)^2 = 4 \sin^2\left(\frac{rx}{2}\right) \cos^2\left(4Ljx + \frac{(2u+r-1)x}{2}\right).$$

We average over the qL^2 triples. For fixed r , and as j ranges over $\{0, \dots, q-1\}$ and u over $\{0, \dots, L-1\}$, the phase $\alpha_{j,u} = 4Ljx + \frac{(2u+r-1)x}{2}$ splits into a box part depending on j and a

shift depending on u . Averaging over j first, the non-resonant phase average gives, for every real α and every $x \notin R$,

$$\frac{1}{q} \sum_{j=0}^{q-1} \cos^2(4Ljx + \alpha) \geq \frac{3}{8}.$$

Applying this with $\alpha = \frac{(2u+r-1)x}{2}$ for each fixed u and r , and then averaging over $u \in \{0, \dots, L-1\}$ and $r \in \{L, \dots, 2L-1\}$,

$$\frac{1}{qL^2} \sum_{j,u,r} Y_{j,u,r}(x)^2 \geq 4 \cdot \frac{3}{8} \cdot \frac{1}{L} \sum_{r=L}^{2L-1} \sin^2\left(\frac{rx}{2}\right).$$

By the contiguous block-length square-mean lemma applied with $t = x/2$ (so that $L \sin t = Ls$),

$$\frac{1}{L} \sum_{r=L}^{2L-1} \sin^2\left(\frac{rx}{2}\right) \geq c_A \min\{(Ls)^2, 1\} = c_A \Delta^2.$$

Combining the two displays,

$$\frac{1}{qL^2} \sum_{j,u,r} Y_{j,u,r}(x)^2 \geq 4 \cdot \frac{3}{8} \cdot c_A \Delta^2 = c_0 \Delta^2, \quad c_0 := \frac{3}{2} c_A > 0. \quad (2)$$

Uniform upper bound. We claim that $|Y_{j,u,r}(x)| \leq 4\Delta$ for every triple. Since the cosine factor in the block-sum identity is bounded by 1,

$$|Y_{j,u,r}(x)| \leq 2|\sin(rx/2)| = 2|\sin(rt)|.$$

If $\Delta = 1$ (that is, $Ls \geq 1$), then $|Y_{j,u,r}(x)| \leq 2 \leq 4 = 4\Delta$. If $\Delta = Ls < 1$, we use the elementary inequality

$$|\sin(rt)| \leq r \sin t \quad (0 \leq t \leq \frac{\pi}{2}, r \in \mathbb{Z}_{\geq 0}). \quad (3)$$

Inequality (3) follows by induction on r : it is trivial for $r = 0$, and for $r \geq 1$ the addition formula together with $|\cos| \leq 1$ and $\sin t \geq 0$ gives

$$|\sin(rt)| = |\sin((r-1)t) \cos t + \cos((r-1)t) \sin t| \leq |\sin((r-1)t)| + \sin t \leq (r-1) \sin t + \sin t = r \sin t.$$

Since $r \leq 2L-1 < 2L$, (3) yields $|\sin(rt)| \leq 2Ls$, whence $|Y_{j,u,r}(x)| \leq 2 \cdot 2Ls = 4Ls = 4\Delta$. In either case

$$|Y_{j,u,r}(x)| \leq 4\Delta. \quad (4)$$

Paley-Zygmund-type deduction. Set $a := \sqrt{c_0/2}$ and let ρ denote the proportion of triples with $|Y_{j,u,r}(x)| \geq a\Delta$. Splitting the average in (2) according to whether $|Y_{j,u,r}| \geq a\Delta$ or not, and bounding the former group by the uniform estimate (4) and the latter by $a\Delta$,

$$c_0 \Delta^2 \leq \frac{1}{qL^2} \sum_{j,u,r} Y_{j,u,r}^2 \leq \rho (4\Delta)^2 + (1-\rho) (a\Delta)^2 = 16\rho \Delta^2 + (1-\rho) a^2 \Delta^2.$$

Since $a^2 = c_0/2$, this reads $c_0 \Delta^2 \leq 16\rho \Delta^2 + \frac{c_0}{2}(1-\rho) \Delta^2$. As $x \notin R \subset \{x : x \in (0, \pi)\}$ and $x \in (0, \pi)$ we have $\Delta > 0$; dividing by Δ^2 and rearranging,

$$c_0 - \frac{c_0}{2} \leq \left(16 - \frac{c_0}{2}\right) \rho, \quad \text{hence} \quad \rho \geq \frac{c_0/2}{16 - c_0/2} =: \gamma > 0.$$

Finally, for each triple counted by ρ we recover a lower bound for $|b_{j,u,r}|$ itself: from $|Y_{j,u,r}| = 2s|b_{j,u,r}| \geq a\Delta$ and $\Delta/s = \min\{Ls, 1\}/s = \min\{L, 1/s\} = A_L(x)$,

$$|b_{j,u,r}(x)| = \frac{|Y_{j,u,r}(x)|}{2s} \geq \frac{a}{2} \cdot \frac{\Delta}{s} = \frac{a}{2} A_L(x).$$

Taking $c_1 := a/2$ completes the proof, with $\gamma = (c_0/2)/(16 - c_0/2)$. $\square$

5.2. Abundance of good blocks. For the random construction, write $b_j(x) = b_{j,U_j,R_j}(x)$ for the block actually selected in box j , where (U_j, R_j) are the independent uniform choices of the block geometry.

Lemma 9 (Abundance of good blocks). *There exist absolute constants $c_2, c_3 > 0$ such that for every $x \in (0, \pi) \setminus R$,*

$$\Pr_{U,R} \left(\#\{j : |b_j(x)| \geq c_1 A_L(x)\} < c_2 q \right) \leq e^{-c_3 q},$$

where c_1 is the constant of Lemma 8.

Proof. Fix $x \in (0, \pi) \setminus R$. For each box j let X_j be the indicator of the event $|b_j(x)| \geq c_1 A_L(x)$, and set $p_j = \mathbb{E}X_j = \Pr_{U_j,R_j}(|b_{j,U_j,R_j}(x)| \geq c_1 A_L(x))$. Since (U_j, R_j) is uniform over the L^2 pairs (u, r) in box j , p_j equals the proportion of pairs (u, r) for which $|b_{j,u,r}(x)| \geq c_1 A_L(x)$. Summing over j and invoking Lemma 8, which counts good triples over all boxes,

$$\sum_{j=0}^{q-1} p_j = \frac{1}{L^2} \#\{(j, u, r) : |b_{j,u,r}(x)| \geq c_1 A_L(x)\} \geq \frac{1}{L^2} \cdot \gamma q L^2 = \gamma q.$$

The variables $X_0, \dots, X_{q-1}$ are independent, being functions of the independent pairs (U_j, R_j) ; they need not be identically distributed, and the argument below uses only independence.

Let $S = \sum_j X_j$ and $\mu = \mathbb{E}S = \sum_j p_j \geq \gamma q$. For $\lambda > 0$, by Markov's inequality applied to $e^{-\lambda S}$ and independence,

$$\Pr(S \leq \mu/2) = \Pr(e^{-\lambda S} \geq e^{-\lambda \mu/2}) \leq e^{\lambda \mu/2} \mathbb{E}e^{-\lambda S} = e^{\lambda \mu/2} \prod_{j=0}^{q-1} (1 - p_j + p_j e^{-\lambda}).$$

Using $1 - p_j + p_j e^{-\lambda} = 1 - (1 - e^{-\lambda})p_j \leq \exp(-(1 - e^{-\lambda})p_j)$ and $\sum_j p_j = \mu$,

$$\Pr(S \leq \mu/2) \leq \exp\left(\frac{\lambda \mu}{2} - (1 - e^{-\lambda})\mu\right).$$

Take $\lambda = \log 2$, so that $e^{-\lambda} = 1/2$ and $1 - e^{-\lambda} = 1/2$, giving

$$\Pr(S \leq \mu/2) \leq \exp\left(\frac{\mu}{2} \log 2 - \frac{\mu}{2}\right) = \exp\left(-\frac{\mu}{2}(1 - \log 2)\right) \leq e^{-\mu/8},$$

where we used $1 - \log 2 > 1/4$. Since $\mu \geq \gamma q$, we obtain $\Pr(S \leq \mu/2) \leq e^{-\gamma q/8}$. Finally $c_2 q \leq \mu/2$ because $\mu/2 \geq \gamma q/2 \geq (\gamma/8)q = c_2 q$, so

$$\Pr(S < c_2 q) \leq \Pr(S \leq \mu/2) \leq e^{-\gamma q/8}.$$

Taking $c_2 = c_3 = \gamma/8$ proves the lemma. $\square$

5.3. A weighted Littlewood–Offord inequality.

Lemma 10 (Weighted Littlewood–Offord, arbitrary centre). *Let $\eta_1, \dots, \eta_r$ be independent Bernoulli(1/2) random variables and let $a_1, \dots, a_r$ be real weights with $|a_i| \geq a > 0$. Then for every $t \in \mathbb{R}$ and every $w \geq 0$,*

$$\Pr\left(\left|\sum_{i=1}^r \eta_i a_i - t\right| \leq w\right) \leq \frac{C(1 + w/a)}{\sqrt{r}},$$

with an absolute constant C .

Proof. Sign reduction. Suppose some $a_i < 0$. Replacing η_i by $1 - \eta_i$ leaves the distribution of η_i unchanged (it is Bernoulli(1/2)), and changes the summand from $\eta_i a_i$ to $(1 - \eta_i)a_i = a_i - \eta_i a_i = \eta_i(-a_i) + a_i$. Thus $\sum_i \eta_i a_i = \sum_i \eta_i a'_i + \sum_{i:a_i < 0} a_i$ in distribution, where $a'_i = |a_i| \geq a$. Absorbing the deterministic shift $\sum_{i:a_i < 0} a_i$ into the centre t , we may and do assume $a_i \geq a > 0$ for all i .

Antichain from a half-open interval. Encode a realization of $(\eta_1, \dots, \eta_r)$ by the subset $S = \{i : \eta_i = 1\} \subseteq \{1, \dots, r\}$, so that $\sum_i \eta_i a_i = \sigma(S) := \sum_{i \in S} a_i$. Fix any half-open interval $J = [\beta, \beta + a)$ of length a . We claim the family

$$\mathcal{F}_J = \{S \subseteq \{1, \dots, r\} : \sigma(S) \in J\}$$

is an antichain under inclusion. Indeed, if $S \subsetneq T$ both lay in J , then $\sigma(T) - \sigma(S) = \sum_{i \in T \setminus S} a_i \geq a$ (the sum being over a nonempty set with each weight $\geq a$), so $\sigma(T) \geq \sigma(S) + a \geq \beta + a$, contradicting $\sigma(T) < \beta + a$. Hence no two distinct members of $\mathcal{F}_J$ are comparable. By Sperner's theorem, $|\mathcal{F}_J| \leq \binom{r}{\lfloor r/2 \rfloor}$, so

$$\Pr(\sigma(S) \in J) = 2^{-r} |\mathcal{F}_J| \leq 2^{-r} \binom{r}{\lfloor r/2 \rfloor} \leq \frac{C}{\sqrt{r}},$$

by the standard central binomial estimate $\binom{r}{\lfloor r/2 \rfloor} \leq C 2^r / \sqrt{r}$.

We stress that the interval must be half-open: for a closed interval of length a the family need not be an antichain. Already for a single weight $a_1 = a$ the interval $[0, a]$ contains both subset sums $\sigma(\emptyset) = 0$ and $\sigma(\{1\}) = a$, and $\emptyset \subsetneq \{1\}$, so these two comparable sets both lie in the closed interval.

Covering the window. The window $[t - w, t + w]$ has length $2w$ and can be covered by $\lceil 2w/a \rceil + 1 \leq 2(1 + w/a)$ half-open intervals of length a together with a single endpoint singleton (a point has probability at most $C/\sqrt{r}$ of being hit, by the case $w = 0$ of the antichain bound). By the union bound,

$$\Pr\left(\left|\sum_i \eta_i a_i - t\right| \leq w\right) \leq (2(1 + w/a) + 1) \cdot \frac{C}{\sqrt{r}} \leq \frac{C'(1 + w/a)}{\sqrt{r}}.$$

Renaming the constant proves the lemma. $\square$

5.4. Pointwise small-ball bound. Recall that $E(g) = \{x \in (0, \pi) : |h(x)| \leq 1\}$ with $h(x) = 2s(g(x) - \frac{1}{2})$, so that

$$x \in E(g) \iff |2s g(x) - s| \leq 1 \iff \left| \sum_{j=0}^{q-1} \eta_j b_j(x) - \frac{1}{2} \right| \leq \frac{1}{2s},$$

after dividing by $2s > 0$ and recalling $g(x) = \sum_j \eta_j b_j(x)$.

Corollary 1 (Pointwise small-ball). *There are absolute constants $C, c > 0$ such that for every fixed $x \in (0, \pi) \setminus R$,*

$$\Pr(x \in E(g)) \leq \begin{cases} C \min\left\{1, (xL\sqrt{q})^{-1} + q^{-1/2}\right\} + e^{-cq}, & 0 < x \leq 1/L, \\ C q^{-1/2} + e^{-cq}, & 1/L < x < \pi. \end{cases}$$

Proof. Fix $x \in (0, \pi) \setminus R$. Condition on the block geometry $(U_j, R_j)_j$, which determines the weights $b_j(x)$ but not the signs η_j ; the latter remain independent Bernoulli(1/2) after conditioning. Let $\mathcal{G}$ denote the good event of Lemma 9, on which the set

$$J = \{j : |b_j(x)| \geq c_1 A_L(x)\}$$

has cardinality $|J| \geq c_2 q$. By Lemma 9, $\Pr(\mathcal{G}^c) \leq e^{-c_3 q}$.

On $\mathcal{G}$ we bound $\Pr(x \in E(g) \mid (U_j, R_j)_j)$ by conditioning further on the signs $\{\eta_j : j \notin J\}$. Writing

$$\sum_{j=0}^{q-1} \eta_j b_j(x) - \frac{1}{2s} = \sum_{j \in J} \eta_j b_j(x) - \left(\frac{1}{2s} - \sum_{j \notin J} \eta_j b_j(x) \right),$$

the second parenthesis is a constant, say $t = t((\eta_j)_{j \notin J})$, once the outside signs are fixed. The remaining randomness is $\{\eta_j : j \in J\}$, independent Bernoulli(1/2), with weights $a_j = b_j(x)$ satisfying $|a_j| \geq c_1 A_L(x) =: a$. Applying Lemma 10 with $r = |J| \geq c_2 q$, centre t , and window $w = 1/(2s)$,

$$\Pr \left(\left| \sum_{j \in J} \eta_j b_j(x) - t \right| \leq \frac{1}{2s} \mid (U_j, R_j)_j, (\eta_j)_{j \notin J} \right) \leq \frac{C(1 + \frac{w}{a})}{\sqrt{|J|}} \leq \frac{C(1 + \frac{w}{a})}{\sqrt{c_2 q}}.$$

Averaging over the outside signs preserves this bound, so on $\mathcal{G}$,

$$\Pr(x \in E(g) \mid (U_j, R_j)_j) \leq \frac{C}{\sqrt{q}} \left(1 + \frac{w}{a} \right), \quad w = \frac{1}{2s}, \quad a = c_1 A_L(x).$$

Window-to-scale ratio. We estimate $w/a = (1/(2s))/(c_1 \min\{L, 1/s\})$. Recall the elementary bounds $x/\pi \leq \sin(x/2) = s \leq x/2$ for $0 < x < \pi$.

If $0 < x \leq 1/L$, then $s \leq x/2 \leq 1/(2L) < 1/L$, so $Ls < 1$ and $\min\{L, 1/s\} = L$. Hence

$$\frac{w}{a} = \frac{1}{2s c_1 L} \leq \frac{1}{2c_1 L} \cdot \frac{\pi}{x} = \frac{\pi}{2c_1} \cdot \frac{1}{Lx} \ll \frac{1}{Lx},$$

using $s \geq x/\pi$. Consequently $1 + w/a \ll 1 + 1/(Lx)$, and since $Lx \leq 1$ in this range the summand 1 is $\ll 1/(Lx)$ as well; thus $1 + w/a \ll 1/(Lx)$. Substituting,

$$\Pr(x \in E(g) \mid (U_j, R_j)_j) \ll \frac{1}{\sqrt{q}} \cdot \frac{1}{Lx} = \frac{1}{xL\sqrt{q}} \quad (0 < x \leq 1/L).$$

If $1/L < x < \pi$, we distinguish two subcases according to the size of s relative to $1/L$.

- If $s \leq 1/L$, then $\min\{L, 1/s\} = L$ and $w/a = 1/(2c_1 Ls)$. From $s > x/\pi > 1/(\pi L)$ (the first inequality by $s \geq x/\pi$ and $x > 1/L$) we get $Ls > 1/\pi$, hence $w/a \leq \pi/(2c_1) \ll 1$.
- If $s > 1/L$, then $\min\{L, 1/s\} = 1/s$ and $w/a = 1/(2c_1) \cdot s/1 = s/(2c_1) \leq 1/(2c_1) \ll 1$, since $s \leq 1$.

In either subcase $w/a \ll 1$, so $1 + w/a \ll 1$ and

$$\Pr(x \in E(g) \mid (U_j, R_j)_j) \ll \frac{1}{\sqrt{q}} \quad (1/L < x < \pi).$$

Conclusion. Combining the good-event bound with the bad event $\mathcal{G}^c$ of probability at most $e^{-c_3 q}$, and taking expectation over the geometry,

$$\Pr(x \in E(g)) \leq \Pr(x \in E(g) \mid \mathcal{G}) + \Pr(\mathcal{G}^c) \leq \begin{cases} \frac{C}{xL\sqrt{q}} + e^{-c_3 q}, & 0 < x \leq 1/L, \\ \frac{C}{\sqrt{q}} + e^{-c_3 q}, & 1/L < x < \pi. \end{cases}$$

In the range $0 < x \leq 1/L$ the conditional Littlewood–Offord bound is also at most $Cq^{-1/2}(1 + w/a)$ with the trivial factor $1 + w/a$, and re-inserting the additive $q^{-1/2}$ arising from the $+1$ in $1 + w/a$ yields the stated form $C \min\{1, (xL\sqrt{q})^{-1} + q^{-1/2}\}$; truncating every probability by 1 and writing $c = c_3$ completes the proof. $\square$

6. THE SMALL-BAND MEASURE ESTIMATE

Throughout this section g denotes the random 0/1 cosine polynomial of the block construction, of degree at most m , and $E(g) = \{x \in (0, \pi) : |h(x)| \leq 1\}$ is its small-band set, where $h(x) = 2 \sin(x/2) (g(x) - \frac{1}{2})$. We recall the parameters $8 \leq L \leq m^{1/3}$ and $q = \lfloor (m+1)/(4L) \rfloor$, and the resonance layer $R = \{x \in [0, \pi] : |\sin(4Lx)| \leq D/q\}$ with $D = 4$; we assume $q \geq 8$, so that $q \geq 2D$.

Proposition 1. *With the notation above,*

$$\mathbb{E} |E(g)| \ll \frac{1}{q} + \frac{\log q}{L\sqrt{q}} + \frac{1}{\sqrt{q}}.$$

Proof. The indicator $\mathbf{1}_{\{x \in E(g)\}}$ is jointly measurable and nonnegative in x and in the randomness $(U_j, R_j, \eta_j)_j$, so Tonelli's theorem gives

$$\mathbb{E} |E(g)| = \mathbb{E} \int_0^\pi \mathbf{1}_{\{x \in E(g)\}} dx = \int_0^\pi \Pr(x \in E(g)) dx.$$

We split the integral according to the resonance layer and, on its complement, according to the scale $x_0 := (L\sqrt{q})^{-1}$.

Contribution of the resonance layer. On R we use only the trivial bound $\Pr(x \in E(g)) \leq 1$. Since $q \geq 2D$, the resonance-measure lemma gives $|R| = 2 \arcsin(D/q) \leq 4D/q$, whence

$$\int_R \Pr(x \in E(g)) dx \leq |R| \leq \frac{4D}{q} \ll \frac{1}{q}.$$

Contribution of the complement. On $(0, \pi) \setminus R$ the pointwise small-ball corollary applies. For $1/L < x < \pi$ it yields $\Pr(x \in E(g)) \leq Cq^{-1/2} + e^{-cq}$, and for $0 < x \leq 1/L$ it yields

$$\Pr(x \in E(g)) \leq C \min\left\{1, (xL\sqrt{q})^{-1} + q^{-1/2}\right\} + e^{-cq}.$$

We treat the three ranges $(0, x_0]$, $(x_0, 1/L)$ and $[1/L, \pi)$ separately; note that $x_0 = (L\sqrt{q})^{-1} \leq 1/L$ since $q \geq 1$.

On $(0, x_0]$ we bound $\Pr(x \in E(g)) \leq 1$ by the truncation in the corollary, so this range contributes at most

$$x_0 = \frac{1}{L\sqrt{q}}.$$

On $(x_0, 1/L)$ we discard the exponential term and the additive $q^{-1/2}$ into the last two ranges below (they are integrated over an interval of length $< 1/L \leq 1$ and $< \pi$ respectively) and retain the dominant factor $(xL\sqrt{q})^{-1}$. Since $x > x_0 = (L\sqrt{q})^{-1}$, we have $xL\sqrt{q} > 1$, so $(xL\sqrt{q})^{-1} < 1$ and the truncation is inactive on this range. Hence

$$\int_{x_0}^{1/L} \frac{C}{xL\sqrt{q}} dx = \frac{C}{L\sqrt{q}} \int_{x_0}^{1/L} \frac{dx}{x} = \frac{C}{L\sqrt{q}} \log \frac{1/L}{x_0} = \frac{C}{L\sqrt{q}} \log(\sqrt{q}) = \frac{C \log \sqrt{q}}{L\sqrt{q}},$$

using $x_0^{-1}/L = \sqrt{q}$. This is $\ll (\log q)/(L\sqrt{q})$.

Finally, on $[1/L, \pi)$ the bound $\Pr(x \in E(g)) \leq Cq^{-1/2} + e^{-cq}$ integrates to at most

$$(\pi - 1/L)(Cq^{-1/2} + e^{-cq}) \ll \frac{1}{\sqrt{q}},$$

since $e^{-cq} \leq Cq^{-1/2}$ for $q \geq 1$; the residual additive $q^{-1/2}$ and exponential terms retained from the range $(x_0, 1/L)$ are of the same order and are absorbed here.

Adding the four contributions gives

$$\mathbb{E} |E(g)| \ll \frac{1}{q} + \frac{1}{L\sqrt{q}} + \frac{\log \sqrt{q}}{L\sqrt{q}} + \frac{1}{\sqrt{q}} \ll \frac{1}{q} + \frac{\log q}{L\sqrt{q}} + \frac{1}{\sqrt{q}},$$

where the term $1/(L\sqrt{q})$ from the range $(0, x_0]$ is absorbed into $(\log q)/(L\sqrt{q})$. This is the asserted bound. $\square$

Corollary 2. *Suppose $q \asymp m/L$ and $L \asymp \log m$; concretely take $L = \lfloor \log m \rfloor$. Then there exists a deterministic 0/1 cosine polynomial g of degree at most m with*

$$|E(g)| \ll \sqrt{\frac{\log m}{m}}.$$

Proof. With $q \asymp m/L$ we have $\sqrt{q} \asymp \sqrt{m/L}$, and Proposition 1 yields

$$\mathbb{E} |E(g)| \ll \frac{L}{m} + \frac{\log(m/L)}{\sqrt{mL}} + \sqrt{\frac{L}{m}},$$

where in the middle term $\log q \asymp \log(m/L)$ and $L\sqrt{q} \asymp L\sqrt{m/L} = \sqrt{mL}$.

Take $L = \lfloor \log m \rfloor$, so $L \asymp \log m$. We bound each of the three terms by $C\sqrt{(\log m)/m}$.

For the first term, $L/m \asymp (\log m)/m$, and since $(\log m)/m \leq \sqrt{(\log m)/m}$ for $m \geq 1$, we have $L/m \ll \sqrt{(\log m)/m}$.

For the third term, $\sqrt{L/m} \asymp \sqrt{(\log m)/m}$ directly.

For the middle term, we use $L \asymp \log m$ to control the numerator: since $1 \leq L \leq \log m$, we have $m/L \geq m/\log m$, and $\log(m/L) = \log m - \log L \leq \log m$, while $\log(m/L) \geq \log m - \log \log m \asymp \log m$; thus $\log(m/L) \asymp \log m \asymp L$. Consequently

$$\frac{\log(m/L)}{\sqrt{mL}} \asymp \frac{L}{\sqrt{mL}} = \sqrt{\frac{L}{m}} \asymp \sqrt{\frac{\log m}{m}}.$$

Here the identification $\log(m/L) \asymp L$ is exactly what makes the middle term no larger than the third; without it the middle term would carry an extra logarithmic factor.

Combining the three estimates gives $\mathbb{E} |E(g)| \ll \sqrt{(\log m)/m}$. Since $|E(g)| \geq 0$, the averaging principle furnishes at least one realization of the random construction such that $|E(g)| \ll \mathbb{E} |E(g)|$; the corresponding g is a deterministic 0/1 cosine polynomial of degree at most m satisfying $|E(g)| \ll \sqrt{(\log m)/m}$. $\square$

7. DETERMINISTIC ZERO COUNTING

Throughout this section g denotes a 0/1 cosine polynomial of degree at most m , and we fix an integer $n > m$. Write $M = n + \frac{1}{2}$ and, following the Dirichlet reduction,

$$h(x) = 2 \sin(x/2) \left(g(x) - \frac{1}{2} \right),$$

so that for $x \notin 2\pi\mathbb{Z}$ the identity $D_n(x) - g(x) = 0$ is equivalent to $\sin(Mx) = h(x)$. Recall the small-band set

$$E(g) = \{x \in (0, \pi) : |h(x)| \leq 1\}.$$

Our aim is the bound $Z(D_n - g) \leq C(n|E(g)| + m + 1)$.

We first record the elementary bound on the number of zeros of a trigonometric polynomial.

Lemma 11 (Zeros of a trigonometric polynomial). *Let $P(x) = \sum_{|k| \leq d} c_k e^{ikx}$ be a trigonometric polynomial with integer frequencies and complex coefficients, not identically zero. Then P has at most $2d$ distinct zeros in any period of length 2π . Likewise, if $P(x) = \sum_{|k| \leq d} c_k e^{i(k+1/2)x}$ has*

half-integer frequencies (complex coefficients, not identically zero), then P has at most $2d$ distinct zeros in any period of length 2π .

Proof. For the integer-frequency case set $z = e^{ix}$. Then $z^d P(x) = \sum_{|k| \leq d} c_k z^{k+d}$ is a polynomial in z of degree at most $2d$, not identically zero because the exponentials e^{ikx} are linearly independent. As x ranges over a period of length 2π , z ranges over the unit circle, so P has at most $2d$ distinct zeros per period.

For the half-integer case, factor out the common half-frequency and reindex by $j = k + d$:

$$P(x) = e^{ix/2} \sum_{|k| \leq d} c_k e^{ikx} = e^{ix/2} w^{-d} \sum_{j=0}^{2d} c_{j-d} w^j, \quad w = e^{ix}.$$

The factor $e^{ix/2} w^{-d}$ is nonzero on the unit circle, so the zeros of P correspond bijectively to the roots of the polynomial $\sum_{j=0}^{2d} c_{j-d} w^j$, of degree at most $2d$. As x runs over a period of length 2π , $w = e^{ix}$ runs once over the unit circle, so P has at most $2d$ distinct zeros per period. $\square$

Theorem 2 (Deterministic zero count). *Let g be a 0/1 cosine polynomial of degree at most m , and let $n > m$. Then*

$$Z(D_n - g) \leq C(n|E(g)| + m + 1),$$

where C is an absolute constant.

Proof. Zeros of $D_n - g$ at points of $2\pi\mathbb{Z}$ contribute a bounded number and are absorbed into the constant, so it suffices to count zeros in $(0, 2\pi)$. By the symmetry established at the end of the proof it is enough to count zeros in $(0, \pi)$ and then double the count, adding only a bounded number of endpoint contributions. On $(0, \pi)$ we have $\sin(x/2) > 0$, so $D_n(x) - g(x) = 0$ is equivalent to $\sin(Mx) = h(x)$. Since $|\sin(Mx)| \leq 1$, no such zero can occur where $|h(x)| > 1$. Set

$$U = \{x \in (0, \pi) : |h(x)| < 1\}.$$

We count the zeros lying in U , then account separately for the finitely many points where $|h(x)| = 1$.

Step 1: the number of components of U . The function h has half-integer frequencies with top frequency at most $m + \frac{1}{2}$, since g has degree at most m and $h(x) = 2\sin(x/2)(g(x) - \frac{1}{2})$. Hence h^2 is an integer-frequency trigonometric polynomial of degree at most $2m + 1$, and so is $h^2 - 1$. It is not identically zero: $h(0) = 0$, so $h^2 - 1$ takes the value -1 at $x = 0$. By Lemma 11 the equation $h(x)^2 = 1$ has at most $2(2m + 1) = O(m + 1)$ solutions in a period. The boundary of U in $(0, \pi)$ consists of such points, so U has at most $O(m + 1)$ connected components, each an open subinterval of $(0, \pi)$.

Step 2: the total variation of the arcsine. On U we have $|h| < 1$, so

$$\alpha(x) := \arcsin h(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

is well defined and continuously differentiable. If $h' \equiv 0$ then h is constant, hence $h \equiv h(0) = 0$ and $\alpha \equiv 0$, so the conclusion of this step is immediate; assume therefore $h' \not\equiv 0$. Now h' is a half-integer-frequency trigonometric polynomial of degree $O(m)$, so by Lemma 11 it has $O(m + 1)$ zeros in a period. Subdivide U at the boundary points of its components and at the zeros of h' ; this partitions U into $O(m + 1)$ open intervals on each of which h' has constant sign, so h is strictly monotone. On such an interval $\alpha = \arcsin h$ is monotone with values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, hence has total variation at most π . Summing over the $O(m + 1)$ intervals,

$$\text{Var}_U(\alpha) = O(m + 1).$$

Step 3: the two lattice-hit families and the critical polynomial. Fix a point $x \in U$ and write $\theta = \arcsin h(x) = \alpha(x)$, so $\sin \theta = h(x)$. The equation $\sin(Mx) = h(x) = \sin \theta$ holds if and only if

$$Mx \equiv \theta \pmod{2\pi} \quad \text{or} \quad Mx \equiv \pi - \theta \pmod{2\pi},$$

that is, if and only if

$$\psi_-(x) := Mx - \alpha(x) \in 2\pi\mathbb{Z} \quad \text{or} \quad \psi_+(x) := Mx + \alpha(x) \in (2\mathbb{Z} + 1)\pi.$$

Thus every zero of $D_n - g$ in U is a lattice hit of ψ_- (into the lattice $2\pi\mathbb{Z}$) or of ψ_+ (into the shifted lattice $(2\mathbb{Z} + 1)\pi$).

We now locate the critical points of $\psi_{\pm}$. Since $\alpha = \arcsin h$, we have $\alpha' = h'/\sqrt{1-h^2}$ on U , and hence

$$\psi'_{\pm}(x) = M \pm \alpha'(x) = M \pm \frac{h'(x)}{\sqrt{1-h(x)^2}}.$$

A point of U is critical for ψ_- or for ψ_+ exactly when $M\sqrt{1-h^2} = \pm h'$, equivalently when $M^2(1-h^2) = (h')^2$, that is when

$$P(x) := M^2(1-h(x)^2) - h'(x)^2 = 0.$$

The function P is an integer-frequency trigonometric polynomial of degree at most $2m+1$: indeed h^2 has degree at most $2m+1$, and h' is half-integer of degree $O(m)$, so $(h')^2$ is integer-frequency of degree at most $2m+1$.

We check $P \not\equiv 0$ by evaluating at $x = 0$. Differentiating $h(x) = 2\sin(x/2)(g(x) - \frac{1}{2})$ and using $\sin 0 = 0$, $\cos 0 = 1$ gives

$$h'(0) = \cos(0) \left(g(0) - \frac{1}{2}\right) = g(0) - \frac{1}{2}.$$

Since the coefficients of g lie in $\{0, 1\}$ and g has degree at most m , we have $|g(0) - \frac{1}{2}| \leq m + \frac{1}{2}$, whence

$$|h'(0)| = |g(0) - \frac{1}{2}| \leq m + \frac{1}{2} < n + \frac{1}{2} = M,$$

the strict inequality being exactly the hypothesis $n > m$. With $h(0) = 0$ this yields

$$P(0) = M^2 \cdot 1 - h'(0)^2 = M^2 - h'(0)^2 > 0,$$

so $P \not\equiv 0$. By Lemma 11, the equation $P = 0$ has at most $2(2m+1) = O(m+1)$ solutions in a period.

Step 4: monotone lattice-hit counting. A strictly monotone continuously differentiable function meets any arithmetic lattice of spacing 2π at most $1 + \text{Var}/(2\pi)$ times, where Var is its total variation across the interval of monotonicity; the same holds for the shifted lattice $(2\mathbb{Z} + 1)\pi$ of spacing 2π . Subdivide U at the boundary points of its components and at the zeros of P . By Step 3 this produces $O(m+1)$ open subintervals, and on each of them P has constant sign, so neither ψ_- nor ψ_+ has an interior critical point there; consequently each of ψ_- and ψ_+ is strictly monotone on each subinterval.

On a single such subinterval I , the number of hits of ψ_- into $2\pi\mathbb{Z}$ is at most $1 + \text{Var}_I(\psi_-)/(2\pi)$, and likewise for ψ_+ into $(2\mathbb{Z} + 1)\pi$. Summing over the $O(m+1)$ subintervals, the number of zeros of $D_n - g$ in U is at most

$$O(m+1) + \frac{\text{Var}_U(\psi_-) + \text{Var}_U(\psi_+)}{2\pi}.$$

Since $\psi_{\pm}(x) = Mx \pm \alpha(x)$ and $x \mapsto Mx$ has variation $M|U|$ over the (open) set U of total length $|U|$, we obtain

$$\text{Var}_U(\psi_{\pm}) \leq M|U| + \text{Var}_U(\alpha) = M|U| + O(m+1)$$

by Step 2. As $M = n + \frac{1}{2} \leq n + 1$, it follows that the number of zeros of $D_n - g$ in $U \cap (0, \pi)$ is at most

$$C(n|U| + m + 1).$$

The boundary set $\{x \in (0, \pi) : |h(x)| = 1\}$ has $O(m + 1)$ points by Step 1, and no zero occurs where $|h| > 1$; these boundary points contribute at most $O(m + 1)$ further zeros. Hence the number of zeros of $D_n - g$ in $(0, \pi)$ is at most $C(n|U| + m + 1)$.

Finally, since $D_n(x)$ and $g(x)$ are cosine polynomials, both are invariant under $x \mapsto 2\pi - x$, so $D_n(2\pi - x) - g(2\pi - x) = D_n(x) - g(x)$. Thus the zeros in $(\pi, 2\pi)$ are the reflections of those in $(0, \pi)$, and the count over $(0, 2\pi)$ is at most twice the count over $(0, \pi)$, plus a bounded contribution from $x = \pi$ and the endpoints. Finally, U and $E(g)$ differ only by the finitely many boundary points where $|h| = 1$, so $|U| = |E(g)|$. Combining the above,

$$Z(D_n - g) \leq C(n|E(g)| + m + 1),$$

as claimed. $\square$

Remark 2. The hypothesis $n > m$ is essential. If $n = m$ one may take $g = D_m$, and then $D_n - g \equiv 0$, so the notion of zero count is vacuous; correspondingly the strict inequality $|h'(0)| \leq m + \frac{1}{2} < M$ used to guarantee $P \not\equiv 0$ in Step 3 fails at $n = m$.

8. PROOF OF THE MAIN THEOREM

We assemble the constructions of the previous sections into a bound on $Z(N)$. Throughout, g denotes a fixed 0/1 cosine polynomial furnished by the existence corollary, of degree at most m and with small-band measure $|E(g)| \ll \sqrt{\log m/m}$.

The exact term count. Let K be the number of coefficients of g equal to 1; equivalently, $K = \sum_{j=0}^{q-1} \eta_j R_j$ is the total length of the selected blocks. Each block I_j has length $R_j \leq 2L - 1 < 2L$, and the blocks are pairwise disjoint, so

$$0 \leq K \leq \sum_{j=0}^{q-1} R_j < 2Lq \leq 2L \cdot \frac{m+1}{4L} = \frac{m+1}{2}.$$

Thus $0 \leq K \leq (m+1)/2$.

Given the target N , set

$$n = N + K - 1.$$

The polynomial g is supported in $\{0, \dots, m\}$ with all coefficients in $\{0, 1\}$, and we shall verify below that $n > m$. Granting this, the frequencies of g all lie in $\{0, \dots, m\} \subseteq \{0, \dots, n\}$, and D_n contributes the coefficient 1 at each frequency $0, \dots, n$. Hence $D_n - g$ has all coefficients in $\{0, 1\}$, and the number of nonzero coefficients is

$$(n+1) - K = (N+K) - K = N.$$

Writing $D_n - g = \sum_{a \in A} \cos(ax)$ with $A \subseteq \mathbb{Z}_{\geq 0}$ and $|A| = N$, we conclude that

$$Z(N) \leq Z(D_n - g).$$

Choice of parameters. Fix

$$m = \lceil N^{2/3}(\log N)^{1/3} \rceil, \quad L = \lfloor \log m \rfloor.$$

As $N \rightarrow \infty$,

$$\frac{m}{N} = N^{-1/3}(\log N)^{1/3} (1 + o(1)) \rightarrow 0,$$

so for all sufficiently large N we have $m < N - 1$. Since $K \geq 0$,

$$n = N + K - 1 \geq N - 1 > m,$$

which confirms the hypothesis $n > m$ needed above (and, in particular, that $D_n - g$ is not identically zero). On the other hand $K \leq (m + 1)/2 \leq N$ for large N , whence

$$n = N + K - 1 \leq N + \frac{m + 1}{2} - 1 \leq 2N.$$

Thus $n \asymp N$.

The chosen L satisfies $8 \leq L \leq m^{1/3}$ for large N , so the construction of the random block polynomial applies with $q = \lfloor (m + 1)/(4L) \rfloor \asymp m/L$; in particular $L \asymp \log m$, and the existence corollary yields a deterministic g with

$$|E(g)| \ll \sqrt{\frac{\log m}{m}}.$$

Conclusion. Apply the deterministic zero-counting theorem with this g and with $n > m$, $n \asymp N$. It gives

$$Z(D_n - g) \leq C(n|E(g)| + m + 1) \ll n\sqrt{\frac{\log m}{m}} + m + 1.$$

Since $n \asymp N$ and $m \asymp N^{2/3}(\log N)^{1/3}$, the first term is

$$n\sqrt{\frac{\log m}{m}} \ll N \cdot \left(\frac{\log m}{m}\right)^{1/2}.$$

Here $\log m \asymp \log N$ and $m \asymp N^{2/3}(\log N)^{1/3}$, so

$$\frac{\log m}{m} \asymp \frac{\log N}{N^{2/3}(\log N)^{1/3}} = \frac{(\log N)^{2/3}}{N^{2/3}},$$

and therefore

$$N \left(\frac{\log m}{m}\right)^{1/2} \asymp N \cdot \frac{(\log N)^{1/3}}{N^{1/3}} = N^{2/3}(\log N)^{1/3}.$$

The remaining term satisfies $m + 1 \ll N^{2/3}(\log N)^{1/3}$ by the choice of m . Combining,

$$Z(N) \leq Z(D_n - g) \ll N^{2/3}(\log N)^{1/3}.$$

This holds for all sufficiently large N , which is precisely the assertion of Theorem 1. $\square$

9. CONCLUDING REMARKS

The bound of Theorem 1 improves the previous estimate $Z(N) \ll N^{2/3}(\log N)^{2/3}$ by a factor $(\log N)^{1/3}$. The saving originates in the balance between the two terms of the deterministic count $C(n|E(g)| + m + 1)$: the degree m is optimized against the expected small-band measure $|E(g)|$, and the logarithmic factor in the latter is what the block length $L \asymp \log m$ is tuned to control.

Two features of the argument delimit its reach. First, the small-ball input is a Littlewood–Offord bound of order $r^{-1/2}$ in the effective number $r \asymp c_2 q$ of good blocks; this square-root barrier propagates into the $q^{-1/2}$ term of the pointwise estimate and, after integration, into the exponent $2/3$. Any improvement of the exponent would seem to require either a stronger

anticoncentration estimate exploiting the arithmetic of the frequency blocks, or a construction in which the good weights are more uniform in size. Second, the resonance layer R , though of negligible measure $O(1/q)$, must be excised because the per-block lower bound degenerates near the resonant frequencies; a treatment retaining R would need a genuinely different lower bound there.

It remains open whether the true order of $Z(N)$ is $N^{2/3}$ up to a power of $\log N$, or whether the logarithmic factor can be removed entirely. A matching lower bound of the form $Z(N) \gg N^{2/3}$ is not known, and the gap between the constructions above and the best available lower bounds is the natural focus for further work.

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